

Task 3b – Evaluation

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Project Aims

- Provide information about the producers, their methods and the benefits of buying locally.
- Allow customers to:
 - View products with transparent pricing and availability
 - Place orders for collection or delivery
 - Manage their accounts and view order history
- Provide a dashboard to enable producers to manage stock levels, update product details and view orders
- Have order management, scheduling and tracking features
- Have a loyalty scheme that provides discounts and special offers to returning users
- Provide accessibility features to support a wide range of users

Project Outcomes and Overview

I have successfully produced a basic functional prototype based on the Clients requests with appropriate CSS styling, including related images, graphics and information. Users are able to register accounts with no issues, and can then later use these credentials to sign in and use the website, when signed in a user is able to carry out a range of activities such as, viewing products and product details, and placing orders for collection or delivery. When active on the website, a user has the ability to view and edit their account information, as well as having an overview of all their order history and their status. Overall, my solution has met most of the Clients requests successfully, with some issues in areas that I plan to work on, on a whole, the solution is functional as a basic Ecommerce website, with some added implementations such as a google login system and complex basket function.

Overall, I think that my prototype was a successful display of my skills, as it met the requirements well with working features and appropriate assets. Based on the feedback I had received, users feel that the website is easy to use, helpful, and straight-forward with useful functions and necessary adjustments. My solution is quite accessible to users, especially with the added accessibility features, as it is simple to navigate and process are easy to execute. However, I feel that there are many improvements that could be made in future to improve the experience for the user, as I will mention in the following sections.

Assets and Content

I gathered most of my assets from FreePik, a website that holds a collection of free to use, often AI generated, images, any other assets were also free-to-use without copyright restrictions. The assets used were all meaningful, and purposeful, they all had correlation to certain functions and products within the application, or were images used on the homepage to give users an idea of the websites purpose, and intent when they first land on the website. On the homepage I used an image of growing crops as a background for my main header to draw focus to the website's slogan and general overview. I wanted it to be clear that the website is all about sustainability and local, ethical farming which I personally think I have highlighted well via the use of images on the homepage.

All content on the pages was thought of by myself after doing research on commonly used slogans and catchy marketing punch-lines. I had also used AI here to give myself similar ideas to big brands and to generate pre-made text for reference. For information about the benefits of local farming and supporting businesses, I had carried out my own research as I worked on creating the solution to ensure I included accurate information from reputable sources. I referenced information from the Gemini AI overview on the search page to aid my research, so that I could include the most reliable, up-to-date, truthful information possible. I think that all of my information carries across a point and makes it clear that the Greenfield solution is here to help with all of your needs regarding local, sustainable produce.

Functional Requirements & User Acceptance Criteria

The solution must allow users to register an account.

The solution meets this requirement completely, users are able to create an account via the registration function with ease. There are options for both individual accounts as well as a google login integration, allowing users a range of methods to create an account for use within the application. However, to further improve this feature in the future I could work on implementing Microsoft account integration to allow users for a larger variety of register options, this could open my application to a broader range of users and would also make the use of the website more convenient for those with Microsoft accounts.

The solution must allow users to sign into their account after registering.

I have successfully met this requirement with my prototype, there are no issues with users using the credentials used to register an account to sign in on my website, users are able to login smoothly with minimal errors that are not from the users own faults. The sign in process also includes helpful error messages to guide users in correcting errors they may have encountered, such as making spelling mistakes, entering invalid credentials, or leaving fields blank. Similar to above, I have also integrated a Google login function to make the process simpler for returning users, allowing them to login with a simple click if this was their chosen method prior, expanding on this, I could improve on this feature even more if I included a function to sign in using Microsoft accounts, as mentioned above, this would broaden the range of users that would be able to use my application as it makes signing in simple.

The solution must allow users to manage their accounts.

When using my solution, users are able to access their account page via the link on the navigation menu, from here they can edit their general account information such as their Email and Password, as well as adding information of their choice such as their Phone Number, 2 factor Authentication, or an external login. The user also has the option to download or delete their data stored on the site to permanently remove their account, this is in compliance with the GDPR and Data Protection Act, as a user should have the following rights: The Right to Access (Request a copy of the personal data we hold about you), and The Right to Deletion (The opportunity to delete your personal data, subject to legal obligations). However, some improvements include the ability to add a Home Address or save bank details in order to pay on the website rather than in person, both of these choices would make the website more convenient for the user and entice users to use the website frequently.

The solution must allow users to view order history.

Users have the ability to view their order history easily via a link on the Navigation menu, order details are displayed in a card format for readability and simplicity, this format allows for easy data management and order tracking for the user. The user can view details regarding the Date when the Order was Created, Order Status, Order Type, and Cost. The user is also able to view more details such as the items included in the order, as well as the quantity, individual price, price per count and total cost of these items. To improve this, I could allow the user to search or filter for certain orders, to see information such as complete orders, pending orders, or orders containing certain products. It may also be useful to allow users to cancel or delete orders to clear up cache storage and reduce visual clutter.

The solution must provide suppliers with a dashboard area to manage products.

Unfortunately, I have not fully met this requirement as I have not created an individual area for Suppliers, however, I have edited the Products and Orders view to create a specialised experience for Suppliers. Suppliers are able to manage, create, edit and delete products via the products page whereas standard users cannot utilise these features. As well as this, suppliers are able to view all users order containing a product from their listing, this allows them to have an overview of customer activity with their brand and can give a better opportunity with regards data analysis as suppliers can view all of their data in a centralised location. To improve my solution, I should work on adding these features into a centralised dashboard for supplier access, this will allow suppliers to manage product stock, product information, and customer orders from one location and can help to reduce confusion and allow for faster workflow for the user.

The solution must have a loyalty scheme that provides discounts and special offers to returning customers.

The solution includes a functional Loyalty Scheme which provides users with a 10% discount after they have completed over 5 orders, I have included a Loyalty Summary on the Basket page in the order summary card, detailing the total amount of money subtracted from the users order once the loyalty discount is applied. An improvement I could make is loyalty checkpoints with rewards that get increasingly better as the user places orders through the website, another improvement could be to view your loyalty progress via a tab in the accounts view, this could help to motivate users to reach specific checkpoints when using the site, increasing the potential number of returning users.

The solution must provide information about the supplier, their methods, and the benefits of buying locally.

Users are able to view information about the top local suppliers in collaboration with Greenfield Local Hub, displaying a general description, the Supplier name, and the contact information for the supplier. I could improve on this by adding further details, such as the products listed by the supplier, the number of orders placed with the supplier and potentially the suppliers ranking in the industry.

The solution must allow users to view products in detail with transparent pricing and availability.

On the solution, users are able to view a list of available products via a card format on the Products page from the navigation menu. The cards display all necessary information about the product such as the product Name, product Price, and current Stock, from here the user is able to add the product to their basket (providing it is available) or view further details including the Supplier that has uploaded the item. To improve this feature, I could add a description of the item onto the details page, and more preview image to give the user more information about the item, which may convince them to purchase it.

The solution must allow users to place orders for delivery or collection.

When placing an order, my solution offers support for both delivery and collection orders. I have allowed the user the choice of Order methods, between delivery and collection, as well as sub-choices such as Delivery Type (Next Day, First Class, and Standard delivery) and Collection Date. I have ensured users cannot select a Collection Date that is in the past as this would be an invalid choice, I have also made it so that users can only select a date that is a minimum of 2 days after the current date, to allow the Suppliers time to prepare the order. To improve this function further, I could prevent users from selecting dates that are far into the future, as fresh produce does not remain in date for a long time so there is no way for a Supplier to guarantee that a product is still available after a certain date, the limit for Collections could be within a 2-week window, to prevent confusion and reduce customer disappointment when items are no longer in stock or not in production when the user's collection date arrives.

The solution must provide a selection of accessibility features to support a wide range of users.

I have somewhat managed to meet this requirement as my web application does contain a range of accessibility support options, that are somewhat functional. The main issue I seem to have had with meeting this requirement is the lack of complete functionality with regards the CSS when certain settings are applied, the main issue I have encountered via Feedback I received is the issue with text scaling for some information. It is important that all accessibility settings function fully, correctly, and as the user requires, otherwise some users may struggle to use the information and access certain important information included in the solution. In future, I aim to fix all of the issues with my CSS to ensure that all settings are completely functional with no issues to ensure my application remain accessible for all types of users.

Non-Functional Requirements & KPI

Performance

I have successfully met this requirement, as all pages included in the solution load within a sufficient timeframe, there is minimal delay and no buffering and all pages load fully. All content provided on the application also has fast load speeds, and there are no missing image files so all images load in, with high quality and accuracy. The lack of performance drops keeps the user interested, it reduces the likelihood of a user exiting the site prematurely due to becoming impatient from slow load times and buffering. Meeting this requirement means that my solution complies with industry standards, making it able to compete with other similar applications.

Security

The solution utilises secure data protection methods, all user data and login information are stored as encrypted files on the database, using a secure password hashing algorithm to ensure data security. The solution also includes role-based authentication, as well as account authorisation, by only allowing users to access their individual data, the solution is complying with the Data Protection Act and GDPR, as all data is kept private and confidential. The risk of user data getting leaked by unauthorised personnel is largely reduced as this data cannot be accessed by anyone, only those with the specified account role, and matching details.

Scalability

The code within my Solution is easy to manage, as it contains a broad range of commenting to allow for third-party collaborators to make updates and aid with development. The solution utilises up-to-date coding conventions, such as proper indentation, and appropriate naming, these precautions allow for fast and efficient updates to be completed by others, who may not be familiar with the code. I have also included a Read Me file in the solution folder containing the steps a developer must take before they launch the application, as well as all functional logins for testing, this reduces confused when new developers who are less familiar with the solution undergo testing or build on the applications code as it gives them a background of the correct steps, they must take for the solution to run. To improve the scalability however, from the feedback I have received, I should increase the amount of commenting included in the solution files, to ensure it is clear exactly what the process of the code means, and the logic behind it.

Capacity

I would say that the solution is able to handle a large number of users while upkeeping high performance, and remaining reliable. Although, I have not been able to test if the solution fully meets this requirement, I am sure that it will be able to withstand a heavy load while remaining responsive as it is crafted using multiple of the built-in functions from MVC, which must follow industry standards. I would not be able to make a fair comment on the capacity of the application as of now, but from what I understand, the application should manage to handle a high volume of registered accounts with increasing web traffic.

Reliability

All information I have included on the web application is up-to-date and retrieved from reputable sources. The resources I have used to gather information are reliable, and credible. I plan to check information regularly to maintain the validity of information I provide, this will help to maintain the websites image, keeping the reputation from becoming damaged due to misinformation.

Usability

The navigation menu on the website is built in consideration of both functionality and ease of use, the buttons on the navigation bar are fully functional if the user is logged in, with only minimal issues if the user is not. The navbar buttons always return the correct endpoint with minimal delay. Users are able to visit their intended destination faster, without issues, which reduces the chance of a user becoming uninterested in the site due to struggles with navigation. As well as this, the user is able to follow any other links (buttons, quick links, etc.) without issues, all links also direct the user to the correct location, increasing the sites usability as it allows the site to forward users to information relevant to their intent. These features help with improving user

experience and heighten the value provided by the website. However, it would be useful in future, if I were to include error endpoints when a user attempts to visit an area they are not authorised for, this would aid in reducing confusion when using the website.

Future Improvements

In future, I should work on making my application more accessible generally, ensuring all accessibility settings are fully functional, for example; enabling text scaling for ALL applicable information. As well as repairing any broken CSS or code functionality mentioned previously in my feedback. I plan to improve my colour scheme and implement more, unique styling to captivate the user and better bring attention to the different aspects of the website, to further grab the user's attention I could add more distinctive graphics and images relevant to the websites intended purpose. I want the website to look and feel unique whilst also clearly conveying a purpose, I would like users to easily link the platform to the Greenfield Local Hub business while being visually blessed by a clean, innovative UI that still holds an earthy, rustic tone. I could improve the communication within the site, adding more prompts and confirmation messages, as well as better guidance for the user. I feel that these improvements would help to enhance user satisfaction and create a better experience when using the application, whether it be for the first time or regularly.

Some more improvements I could make include, a cookies pop-up to help manage user sessions and gather important information with regards user web activity to help tailor a custom experience for each unique user. I could also consider implementing a Terms & Services menu that is displayed when a user first visits the website, detailing the terms included when utilizing the application. These improvements would aid in creating a more user-friendly, guided interaction, and can entice users to use the application on more than one occasion, as a custom-fit user environment can interest users as they are more likely to view products and information more suited to their needs and interests.